AI Enhanced Complaint Management System for Government Marketplaces A Hybrid NLP and Backend Data Approach

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ARTICLE INFO

Keywords:
Bengali NLP
Complaint Management
Text Classification
Random Forest
TF IDF
Supabase
NityMulya Application
E-Governance

ABSTRACT

Government marketplaces in Bangladesh receive many complaints every day. Most complaints are written in Bengali or mixed Bengali and English, so manual checking is slow and often inconsistent. This paper presents an AI based complaint management pipeline using data from the NityMulya application. We collected 500 complaints and stored them in a Supabase PostgreSQL database. Each record includes complaint text, product category, and backend numeric values such as validity and priority. We used TF IDF text features, one hot encoded product category features, and scaled numeric features in one machine learning pipeline. We handled class imbalance with SMOTE and trained a Random Forest classifier. On the test split (20 percent, 100 complaints), the model achieved 58.00 percent accuracy, weighted precision of 0.59, weighted recall of 0.58, and weighted F1 score of 0.58. The medium class performed best, while weak and high classes overlapped. This system gives a useful baseline and shows clear directions for future improvement.

1. Introduction

Government marketplaces in Bangladesh are very important. They provide daily goods and services to millions of people. When problems happen, citizens need a fast way to submit complaints and get support.

Today, complaint handling is mostly manual. Citizens usually write complaints in Bengali. Government staff must read each complaint, understand it, and choose a category. This takes a lot of time.

Handling thousands of complaints this way creates many problems. Staff cannot manage the full workload. Mistakes happen when people are tired. Citizens become frustrated when they wait many days for a response. Over time, trust in the system drops.

To solve these problems, we built an AI complaint management system. Complaints are collected from citizens through the NityMulya mobile application. The complaint text is analyzed with NLP methods. The system also uses backend numeric values, such as validity and priority, instead of using only text. This hybrid design gives better support for administrators and provides a clear baseline for future model upgrades.

2. Literature Review

Many researchers have worked on text classification and NLP. Their work guided this study.

Arif et al. (2024) studied deep learning for Bengali hate speech detection on social media [1]. Their work helps explain how short and informal Bengali text can be represented for machine learning.

Ahmed et al. (2023) used transformer models for context based emotion detection from Bengali text [2]. They showed that context is important for good prediction. However, transformer models usually need high computing resources.

Ali et al. (2023) compared Google T5 and SpaCy for text summarization [3]. Their analysis shows that choosing the right text processing method is very important.

To handle class imbalance, Chawla et al. (2002) introduced SMOTE [4]. This method creates synthetic minority samples and is widely used to reduce bias toward majority classes.

Traditional feature extraction methods are still useful for many tasks. Trstenjak et al. (2014) evaluated TF IDF for document classification and showed it is a strong baseline [5].

In public administration, Al-Ruaily et al. (2018) highlighted the need for automated grievance systems in e-governance platforms [6].

Unlike many text only deep learning studies, our work uses a hybrid model. We found that short government complaints are hard to classify with text alone. By combining backend numeric values with TF IDF features, we get a stronger and more practical system at lower cost.

3. Data Collection and Analysis

Reliable machine learning needs reliable data. Our dataset was collected from real users.

3.1. NityMulya App and Supabase Integration

Citizens submit complaints through the NityMulya application. The form captures complaint text, product category, and shop information. After submission, data is stored in a Supabase PostgreSQL database.

ORCID(s):

During insertion, the backend automatically calculates numeric fields using predefined logic. It computes validity and priority scores using pricing related rules.

3.2. Dataset Characteristics

A total of 500 real complaints were extracted from the database for this research. Each record has multiple features. The primary features include:

1. Complaint Description: The text written by the user in Bengali, English, or Banglish.
2. Product Category Name: The product category associated with the complaint.
3. Validity: A backend generated numeric score indicating complaint legitimacy.
4. Priority: A backend generated numeric score indicating urgency.
5. Target Classification: The final label in three classes: weak, medium, and high.

Most complaints express dissatisfaction, which is normal for a grievance platform. The label distribution is imbalanced, and this affects model learning. In the test split (100 records), class counts were 14 for weak, 62 for medium, and 24 for high. Supabase backend fields keep numeric values consistent with marketplace rules, but text still overlaps across classes.

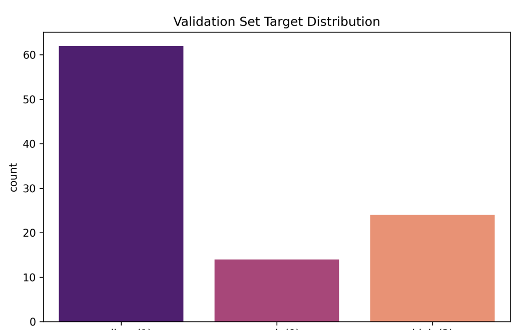

Figure 1: Distribution of evaluation samples in the test set generated by the training pipeline.

4. System Architecture

The processing pipeline was built with Python and Scikit Learn. It converts raw complaint records into model predictions.

4.1. Feature Engineering

Text cannot be used directly by Random Forest. Complaint text was converted to numeric vectors using TF IDF. We used a Feature Union with word and character level features. This helps the model handle spelling variation and mixed language text.

The numeric features validity and priority were processed with StandardScaler. This puts values on a comparable scale.

Table 1
Classification performance from the model training results

Class	Precision	Recall	F1 score	Support
Weak	0.31	0.29	0.30	14
Medium	0.70	0.63	0.66	62
High	0.48	0.62	0.55	24
Weighted Avg	0.59	0.58	0.58	100

Product category is a categorical feature. We used OneHotEncoder to convert product category names into machine readable binary vectors.

4.2. Data Balancing and Training

In real data, some classes have fewer samples. We used SMOTE to balance the training set without removing records.

Finally, the processed data was trained with RandomForestClassifier. Random Forest combines many decision trees and gives stable results.

4.3. Implementation Details

The main experiment was run from the model training workspace using the pipeline script. The workflow includes data loading, preprocessing, feature building, SMOTE balancing, model training, and automatic report generation. Output files include a classification report and visual charts: confusion matrix, test distribution, PCA clusters, and K Means PCA clusters.

5. Results and Analysis

We evaluated the system on a hold out test split of 20 percent from the 500 complaint dataset. This gives 100 records for final testing.

5.1. Model Accuracy

The hybrid Random Forest model achieved 58.00 percent accuracy. Weighted precision, weighted recall, and weighted F1 were 0.59, 0.58, and 0.58. This is a useful baseline, but the model still struggles when complaint meanings are very similar.

5.2. Classification Reports

The pipeline generated a detailed classification report. The medium class had the best precision and F1 score because it has the most samples. The weak class had fewer samples and lower separability, so its score is lower. SMOTE helps with class balance, but it cannot fully remove text overlap between classes.

5.3. Confusion Matrix Evaluation

The confusion matrix shows many predictions around the medium class. There are also errors between weak and high severity boundaries. This means current text signals are not strong enough to clearly separate all classes.

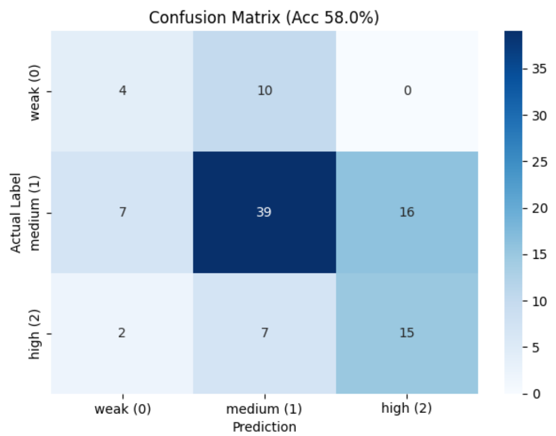

Figure 2: Confusion matrix generated from the test split.

5.4. Feature Space and Cluster Analysis

To better understand class separation, we analyzed PCA cluster plots from the experiment outputs. The plots show partial grouping with visible overlap between classes. This explains the moderate class wise performance.

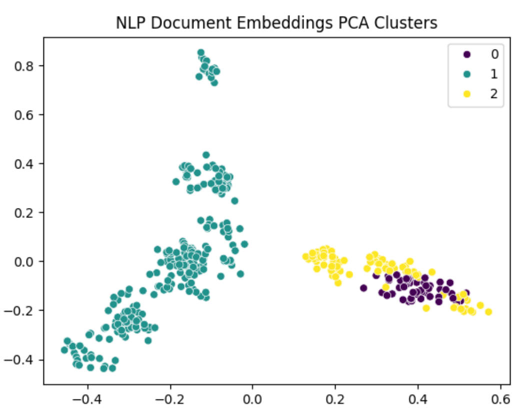

Figure 3: PCA view of transformed complaint feature space.

The current pipeline is still practical for deployment because it is lightweight and fast on CPU systems. For better accuracy, future work should collect more complaints, improve label quality, and compare this baseline with Bengali transformer models.

6. Conclusion

We built an AI complaint management system to classify user issues on the NityMulya platform. By combining TF IDF text features with Supabase backend numeric features, we created a practical Random Forest baseline on 500 verified complaints. The model achieved 58.00 percent test accuracy. It performs better on the majority class, while minority class confusion is still a challenge. Even so, this workflow is reproducible, lightweight, and useful as a strong base for future improvements with more data, better labels, and advanced Bengali language models.

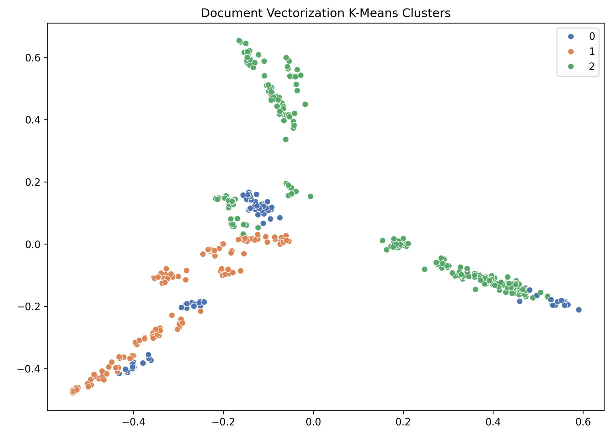

Figure 4: K Means cluster projection over PCA components.

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